

RELIABILITY: A NEW APPROACH IN DESIGN OF INVERTERS FOR PV SYSTEMS

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Abstract— Several surveys of single-phase inverters for residential installations have been reported recently. However, the surveys are aimed at establishing a classification of the topologies reported, and do not explicitly take into account the reliability. This paper presents a reliability-oriented survey of the inverter topologies reported in the literature. The survey is aimed at finding out how well the designers have solved the reliability issue. As such, it complements previous surveys focused at issues such as maturity of the technology. Also a reliability prediction of four photovoltaic systems is presented: a two-stages system, a three-stages one, and two integrated topologies, one with a boost-inverter, and the second with a buck-boost inverter. The goal is to identify the most failure prone components, and the stress factors with the highest contribution to the failure rate. In all cases, it was found that the transistors are the most vulnerable components, and that the dominant stress factor is related to the temperature. It was also found that the reliability can be improved if the switching devices are overrated, but only to a certain level, and using too large transistors can be counterproductive

I. INTRODUCTION

There is currently a widespread installation of residential single-phase PV systems, with powers that typically range from 1 kW to about 5 kW. A major cause of concern is the reliability of the inverter motivated, in part, by the fact that a PV cell has an operational life that exceeds a 20 years span. An inverter usually exhibits a much shorter operational life, and there is an effort to design and manufacture high reliability inverters, with 10 years mean time between failures [1][2][3]. This goal is quite challenging to designers, who must design power stages with reliability in mind.

There are a large number of options available for the inverter in a PV system, ranging from the well-known buck inverter, to integrated topologies. Several surveys of single-phase inverter aimed at residential installations have been reported recently [4][5][6][7][8][9]. However, the surveys are aimed at establishing a classification of the topologies reported, and do not explicitly take into account the reliability. Only one [9] attempts to describe this subject, albeit in an incomplete manner, establishing as a guideline the number of active devices in the power stage. This approach, however, does not provide a clear insight of the reliability attained by the topology, because both the number of devices and its switching mode have an impact on the reliability.

Recently, a methodology to upgrade the converter reliability has been presented. It is not specifically aimed at PV systems, but to power converters in general. According to the guidelines provided, several factors (duty cycle limits, isolation requirements, EMI, switching mode, continuous or discontinuous conduction mode, etc.) should be taken into account during the design cycle [10][11]. Selecting a suitable topology is the first step toward a high reliability converter. This paper is a step toward identifying the most reliable topologies for single-phase PV systems. It presents the reliability-based analysis of four inverters previously reported in the technical literature: a two-stage PV system, a three-stage one, and two integrated topologies. It includes the calculation of reliability-related parameters, such as the failure rate, following the procedure outlined in MIL-HDBK 217 [12]

II. BACKGROUND

The mean time between failures (MTBF) can be calculated using the procedure outlined in the MIL-HDBK 217. The MTBF is the reciprocal of the failure rate λ . The MIL-HDBK 217 lists the failures rates λ_b for electronic devices; however, to predict the reliability of an electronic assembly, the listed value must be multiplied by the π factors that take into account the stresses (electrical, thermal, etc.) on the devices; that is:

$$\lambda = \lambda_b (\pi_a + \pi_b + \pi_c + \dots)$$

On the other hand, the reliability R can be calculated as:

$$R = e^{-\lambda t}$$

Unless some kind of redundancy is included, the reliability R_s of the power stage in a PV system is calculated as:

$$R_s = \prod_{j=1}^n R_j$$

where the term R_j corresponds to the individual reliabilities of the components in the power stage. It follows, then, that a soft-switching topology does not necessarily provide a better reliability that a hard-switching approach, specially if soft-switching is achieved at the expense of a larger number of devices.

III. POWER TOPOLOGIES

For evaluation purposes, the topologies were classified as follows (Fig. 1):

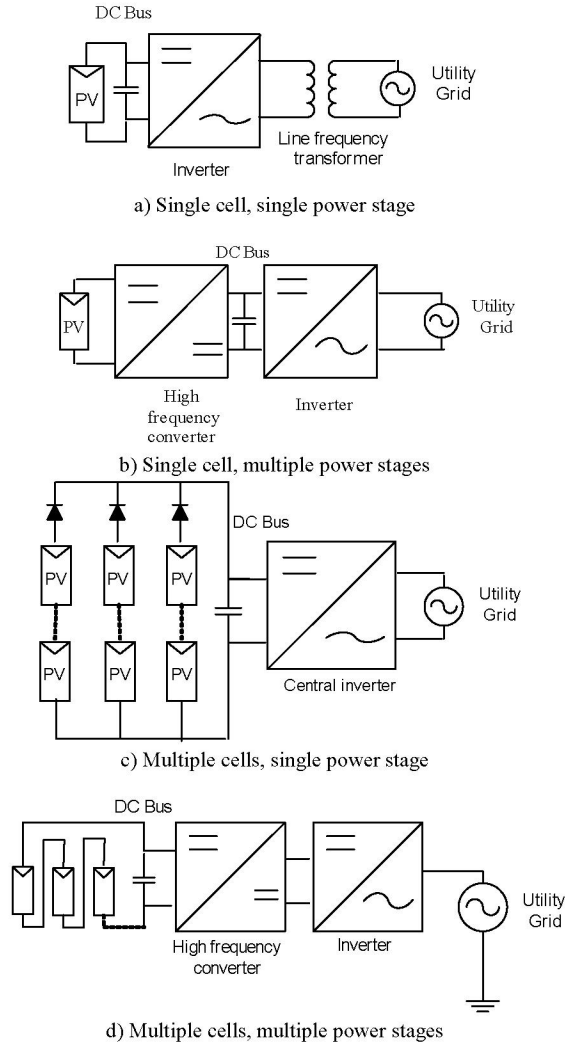


Fig. 1. Classification of PV inverters

- Single cell; single power stage. Power stages with this configuration usually include a small number of components, and exhibit a high efficiency [9].
- Single cell, multiple power stages. There are several options in this category. The most common includes a PWM inverter, and a DC/DC conversion stage. Some examples include ZVS, or a current-fed flyback [5][8][13][14][15][16][17][18][19][20][21][22][23].
- Multiple cells, single power stage. This category includes the centralized inverter, the multilevel inverter and the string inverter [4][7][8][9].
- Multiple cells, multiple power stages. This category includes multi-string converters [8][9][24].

Several configurations were analyzed, and results for fifteen topologies are summarized in table 1. By far, the most common approach followed by designers is aimed at improving the efficiency of the power stage, with 60% of the designs following this approach. However, in many cases the improvements in efficiency have been quite marginal. Both cost and volume seem to receive the same relevance. Modularity comes next, and about 45% of the designs follow this approach. Oddly enough, sturdiness, which can be loosely related to reliability, comes last, and only 33% of the reported designs follow this approach. In fact, less than 15% of the design established this parameter as the main design criterion.

The first topology includes a buck inverter with a small number of components, and a transformer at the mains frequency, that provides galvanic isolation from the grid disturbances, thus increasing the reliability.

Topologies 2 to 5 are aimed at reducing the cost and volume. They include a large number of components. Topologies 6 to 9 are of the integrated type, and are also aimed at cost and volume reduction. They include a small number of components but are intended to be mounted next to the PV cells. Therefore, the environmental stress, and the fact that they lack galvanic isolation, might reduce its reliability.

Topologies 10 and 11 are quite similar, the difference being the number of PV cells that provide the DC voltage. The power stages are simple, with a small number of components, and have good reliability. Topology 13 is aimed at sturdiness and efficiency, but lacks galvanic isolation between the grid and the power stage. Topologies 14 and 15 are quite similar, the difference being the inclusion of a high-frequency transformer between stages, thus increasing the reliability.

IV. PV SYSTEMS COMPARISON

This section presents the analysis of four different configurations. The goal is to identify which parameters (complexity, efficiency, volume, etc) have the greatest impact on reliability.

The two-stages topology is shown in Fig. 2. The first stage is an interleaved boost converter that draws a low-ripple current from the DC source. The second stage is a buck inverter. The system specifications are as follows: $f_{S(\text{Boost})} = 75 \text{ kHz}$, $f_{S(\text{Inverter})} = 5 \text{ kHz}$, $V_{IN} = 170 \text{ V}$, $V_C = 350 \text{ V}$, $V_{ca} = 220 \text{ V @ } 60 \text{ Hz}$. The circuit elements are listed in table II.

TABLE I.
DESIGN APPROACH FOLLOWED IN THE DESIGN OF THE POWER STAGE

No.	Topology	Design approach				
		Sturdiness	Cost	Efficiency	Volume	Modularity
1	Buck Inverter + Frequency line transformer [9]					
2	Modified buck inverter + series resonant converter [16]					
3	Current-fed flyback inverter [17]					
4	Current-fed inverter + boost converter + push-pull converter [18]					
5	Current-fed inverter + current-fed push-pull converter + buck converter [19]					
6	Boost CD-CA converter [20]					
7	Buck-boost inverter [21]					
8	Buck-boost inverter (chopper type) [22]					
9	Dual flyback inverter [23]					
10	Centralized inverter [8]					
11	String inverter [7]					
12	Multilevel inverter [4]					
13	Buck inverter + boost converter [9]					
14	Non-isolated multistring inverter [8]					
15	Isolated multistring inverter [8]					

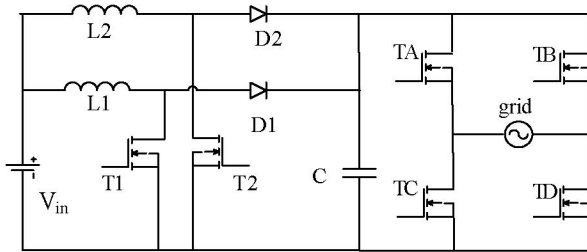


Fig. 2 Two-stages PV system

TABLE II.
CIRCUIT ELEMENTS

Label	Device
T1-T2 TA-TD	SPP04N50C3
D1-D2	15ETX06
L1-L2	6.93 mH
C	1.096 μ F @ 400 V, polypropylene, metalized film

The three-stages PV system is shown in Fig. 3 [18]. The first stage is a boost DC/DC converter. The second stage is another DC/DC converter, this time in the push-pull configuration, that generates a full-wave rectified voltage. The third stage is a bridge inverter that behaves as a polarity inverter, switching at the mains frequency. The system specifications are as follows: $f_s = 100$ kHz, $V_{IN} = 170$ V, $V_{C2} = 200$ V, $V_{CA} = 220$ V @ 50 Hz. The circuit elements are listed in table III.

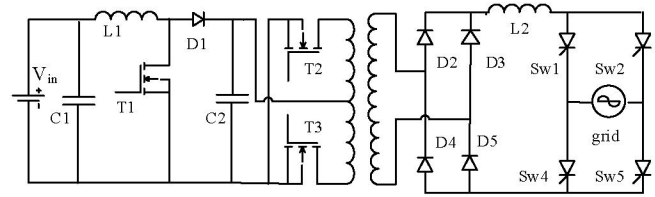


Fig. 3. Three-stages PV system

TABLE III.
CIRCUIT ELEMENTS, AS LISTED IN REFERENCE [14]

Label	Device
T1	IRFP254
T2-T3	IRFP460
D1	BYV29400
D2-D5	BYR29800
Sw1-Sw4	BT152800
L1	68 μ H, RM 10
L2	4 mH E42/15
TR	Transformer, R 36
C1	3.3 μ F @ 250 V, polypropylene, metalized film
C2	550 μ F @380 V, electrolytic, aluminum

An integrated boost inverter [20] is shown in Fig. 4. This is a symmetrical circuit formed by two bi-directional DC/DC converters. Each one generates a DC output, plus an alternating voltage at the mains frequency, and 180° out of phase from each other. The output is obtained in a differential manner, between the outputs, and each converter provides

half the output power. The system specifications are as follows: $P_O = 500\text{W}$, $f_S = 30\text{ kHz}$, $V_{IN} = 48\text{ V}$, $V_{CA} = 220\text{ V @ } 60\text{ Hz}$. The circuit elements are as listed in table IV.

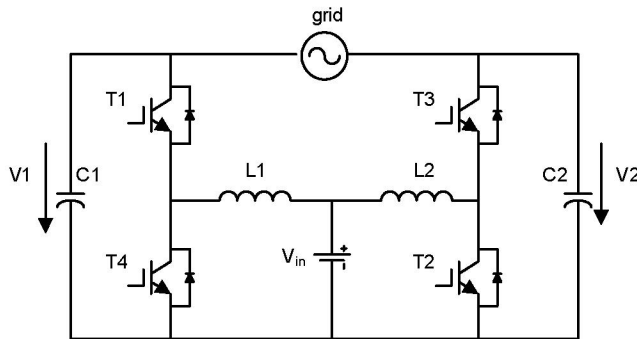


Fig. 4. Boost inverter

TABLE IV
CIRCUIT ELEMENTS, BOOST INVERTER

Label	Device
T1-T4	SPP08N50C3
L1-L2	424.25 μH
C	17.58 $\mu\text{F @ } 250\text{ V}$, electrolytic, aluminum

An alternative to the previous configuration is the buck-boost inverter [21], shown in Fig. 5. It is also a symmetrical circuit with a differential output. The advantage is that the DC level at each output is lower than in the previous circuit. The circuit specifications are as the previous one. The circuit elements are listed in table V.

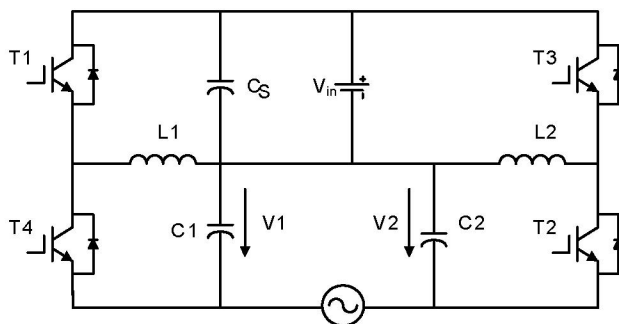


Fig. 5. Buck-boost inverter

TABLE V
CIRCUIT ELEMENTS, BUCK-BOOST INVERTER

Label	Device
T1-T4	SPP08N50C3
L1-L2	424.25 μH
C	22.5 $\mu\text{F @ } 250\text{ V}$, electrolytic, aluminum

To perform a fair comparison among the different topologies, they must be designed for the same operating point, and assume that the operating conditions are similar. Thus, all the photovoltaic systems were designed with a 500 VA output rating. The inverters were designed using the equations included in the papers that described its operation

or, for well-known configurations, the standard procedures described in text books. Once the values of the passive elements were calculated, similar criteria were then applied to select the particular components used in the four designs.

Once the components types and values are known, the failures rates λ_b can be readily obtained from MIL-HDBK 217, but the stress factor must be calculated for the particular application. These factors depend on the maximum voltages and currents in the components; in turn, these parameters were obtained by simulating the power circuits in SPICE and PSIM. The actual reliability calculations were performed using Relx [25], a commercial software package that includes a database with the component failure rates, and executes the procedure in MIL-HDBK 217. It calculates the stress factors when maximum voltage, current, and power dissipation for each component are provided. It also calculates and plots reliability parameters, such as failure rate or MTBF, and its behavior over temperature or time.

It should be noted that the calculations only included the elements in the power stage. The control circuitry and other elements, such as the transistor drivers, were not included in the analysis.

Figures 6 and 7 show the failure rate over temperature, and the MTBF over temperature, for the four circuits analyzed. It can be seen that the two-stages configuration has the highest failure rate. This is because this topology includes six MOSFETs, with fairly high stresses. The other configurations include fewer transistors (3, 4, and 4 respectively). The most complex configuration is the three-stages one; however, in spite of this, it exhibits the best failure rate. This can be explained by two reasons. The first one is that the circuit was simulated with the devices listed in the reference where it is reported, and the transistors are overrated. The second reason is that the third stage is implemented with thyristors, switching at a very low frequency; therefore, this stage has a high reliability.

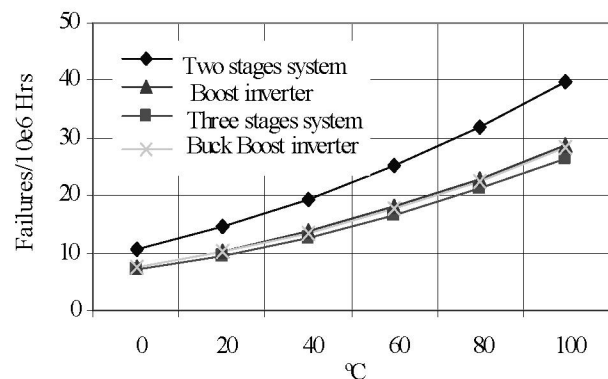


Fig. 6. Failure rate over temperature

For temperatures above 60-70°C, some topologies could have MTBFs lower than 40,000 hours (about 5 years). This is an important issue to ponder when the PV system is to be installed in hot weather areas, such as some areas in Mexico.

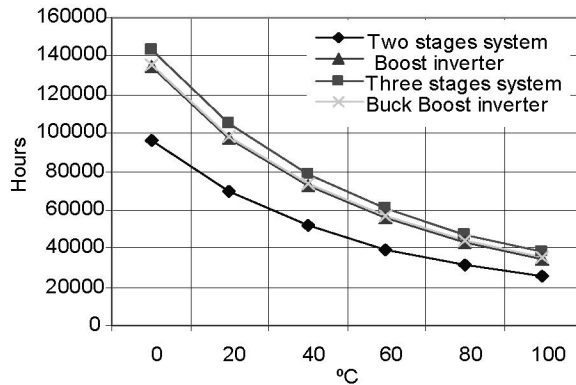


Fig.7. MTBF over temperature

V CONCLUSIONS

From the analysis of the reported topologies, the following conclusions can be drawn:

- Most of the topologies were not developed taking reliability into account. In fact, this issue seems to receive the least attention. Therefore, it is questionable whether the designers are responding to the real requirements of the application, or to the perceived ones.
- When dealing with reliability, the design efforts are aimed at eliminating the electrolytic capacitors. The gain in reliability, however, has not been quantified. This is relevant because some of the proposed solutions are intended to be mounted next to the PV cell, where environmental stresses are higher.
- There is a trend to divide the cells bank in smaller sections, each with a dedicated converter. This might increase the reliability of the entire system, although the converters themselves are not necessarily improved.
- There is a trend to develop converters without transformers at the mains frequency. However, a low-frequency transformer usually provides better immunity to mains disturbances.

From the reliability analysis of the selected topologies, the following conclusions can be drawn:

- The capacitor is not necessarily the weakest link in the circuit. In the four circuits analyzed, the switching transistors were the weakest link. Other components, such as the inductors, do not contribute significantly to the failure rate.
- The complexity of the circuit is not necessarily related to the reliability, although it might become important when some other issues, such as the efficiency, are taken into account.
- The stress factor with the highest contribution to the failure rate is π_T . Therefore, the thermal design must be carefully performed. Overrating the transistors might help increase the reliability, but only to a certain point.
- Reliability can be compromised when the design is

dictated by parameters such as volume (usually involving higher frequency switching, and corresponding higher losses)

It should be noted that the conclusions apply only to the configurations analyzed, the four operating in hard-switching mode. It remains to investigate the reliability behavior of soft-switching configurations, and other, more complex topologies, such as multilevel inverters, that have already been proposed for single-phase grid connected PV systems. In any case, identifying the stress factors with the highest contribution to the failure rate, will always help improve the overall reliability.

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